Management of Bone Injuries & Orthopaedics



Learning Outcomes

- Understand and describe indirect bone fracture healing
- Discuss Implications of orthopaedic approach for physiotherapy management
- Consider the American Orthopaedic Foundation’s Classification of long bone fractures.
- Orthopaedic Surgeries for hip and knee joint
- Physiotherapy Management after hip and knee joint surgery

Why do we care?

- Often in lead up to major competition
- Can (and have) led to complete bone fracture
- Necessitates reduced loading







Video - fractures https://youtu.be/HP6xaEcN6dc



What is a fracture

A fracture is a broken bone. It can range from a thin crack to a complete break. Bone can fracture crosswise, lengthwise, in several places, or into many pieces. Most fractures happen when a bone is impacted by more force or pressure than it can support.

A break in the continuity of the bone, when the capacity of the bone to absorb force is exceeded.

Functions of the skeleton

Hormone Secretion – Osteocalcin – helps to rebuild the skeleton – also helps regulate blood sugar levels and helps burn fat.

1. Support. Bones of the limbs and vertebral column the body; the mandible and maxilla support the teeth; and support some viscera are supported by nearby bones.

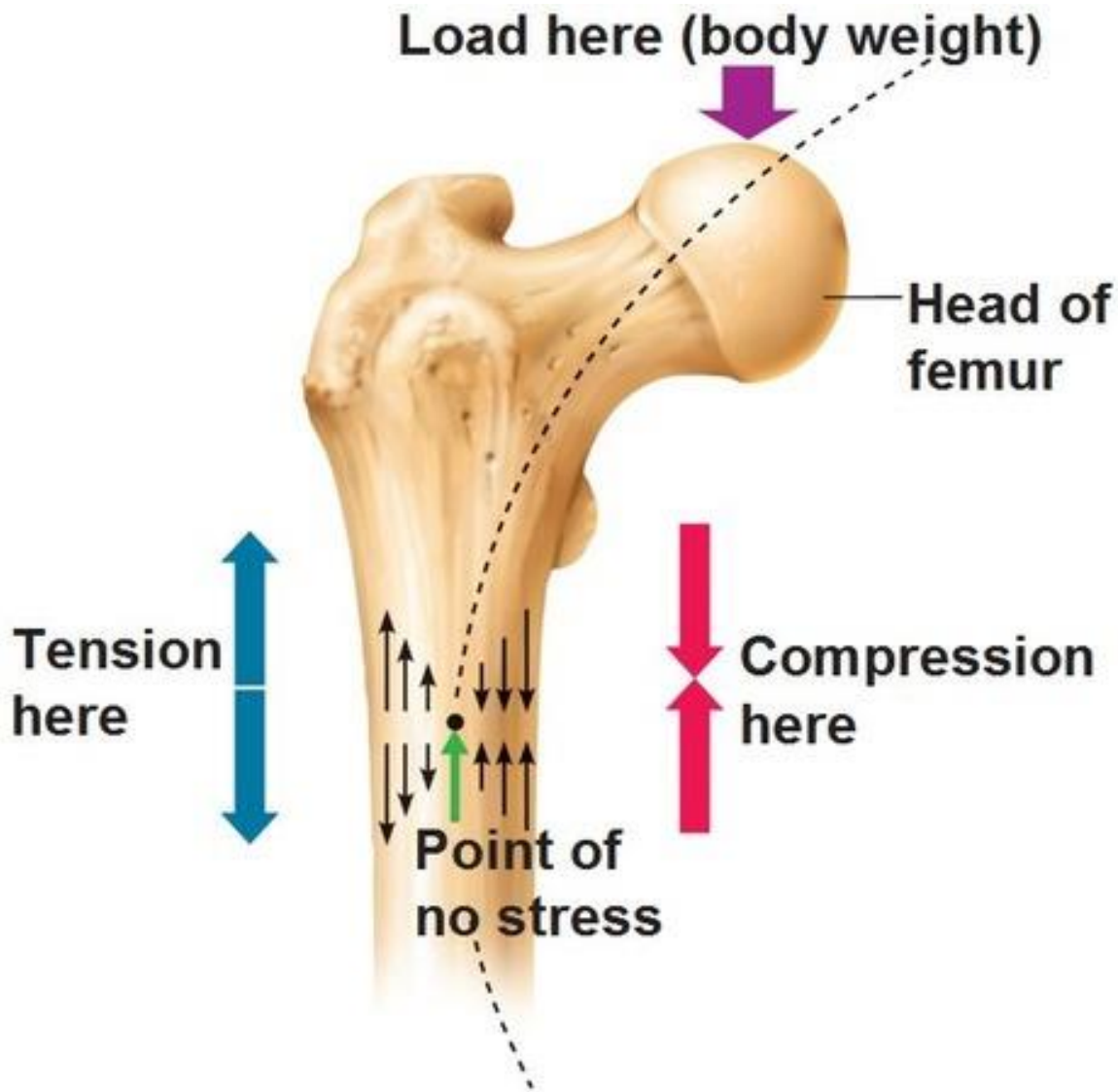
2. Protection. Bones enclose and protect the brain, spinal cord, heart, lungs, pelvic viscera, and bone marrow.

3. Movement. Limb movements, breathing, and other movements are produced by the action of muscles on the bones. Provide Levers.

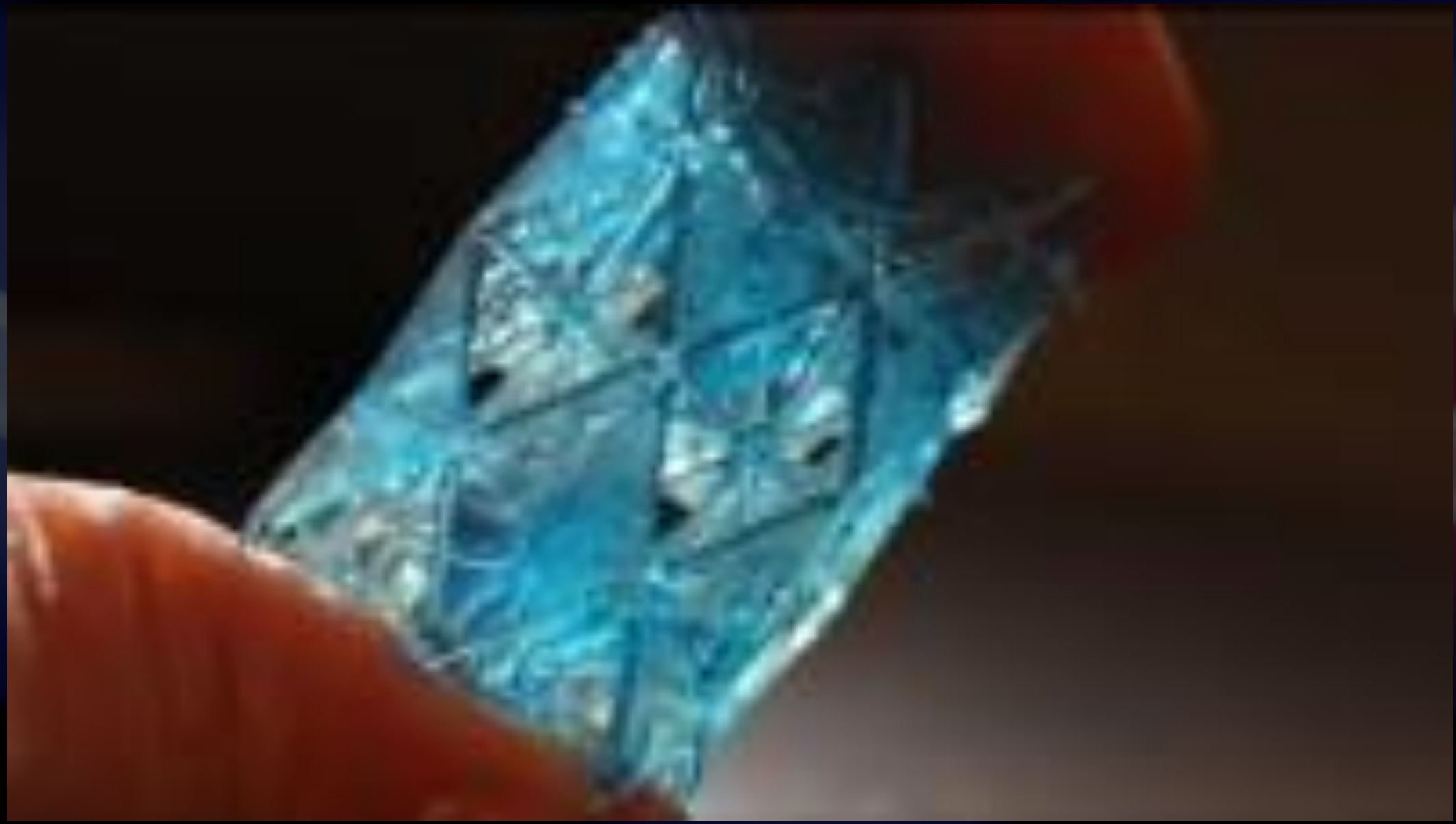
4. Electrolyte balance. The skeleton stores calcium and phosphate ions and releases them into the tissue fluid and blood according to the body's physiological needs.

5. Acid-base balance. Bone tissue buffers the blood against excessive pH changes by absorbing or releasing alkaline phosphate and carbonate salts.

6. Blood formation. Red bone marrow is the major producer of blood cells, including cells of the immune system.



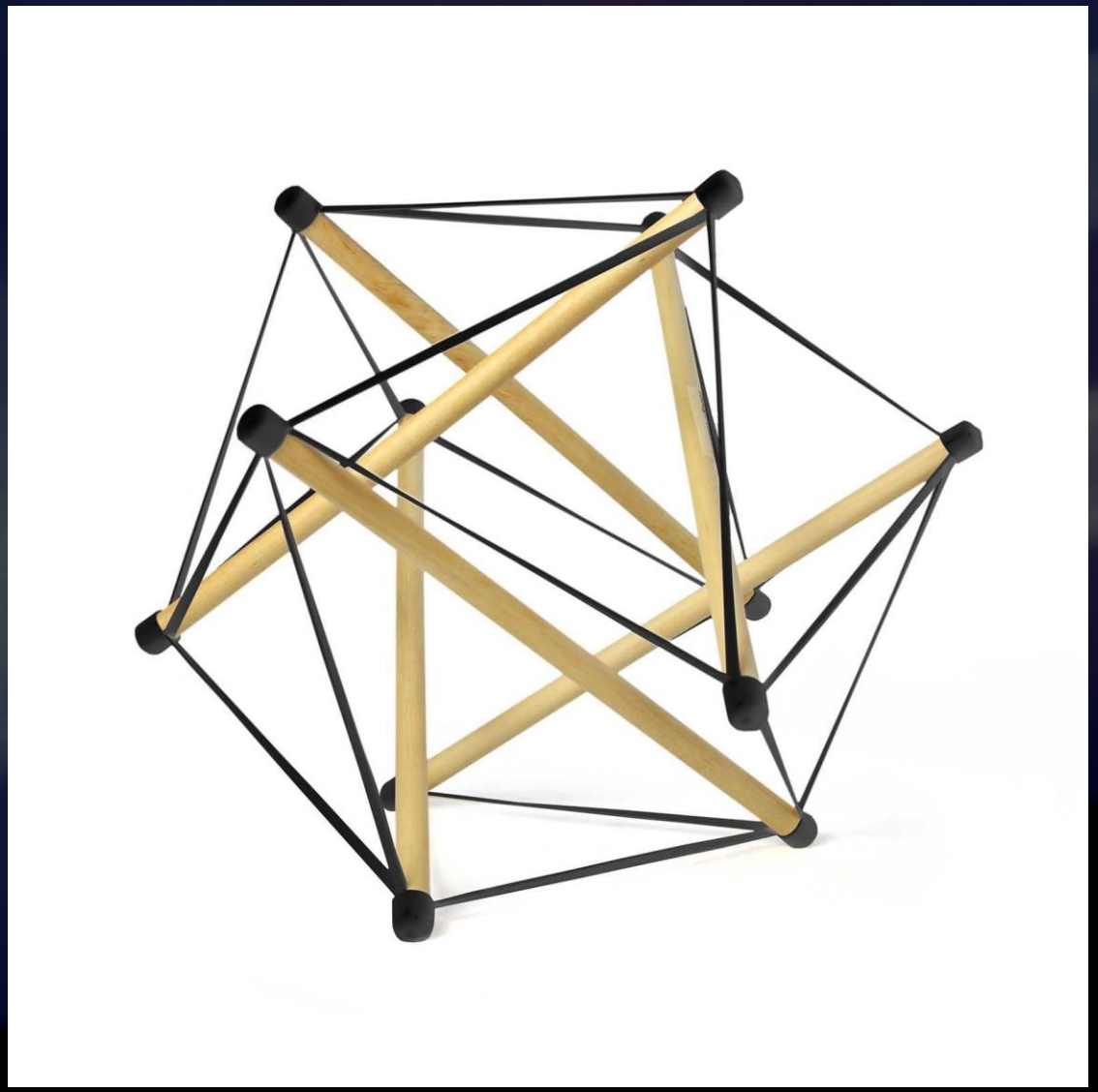
Do bones experience compression or tension?

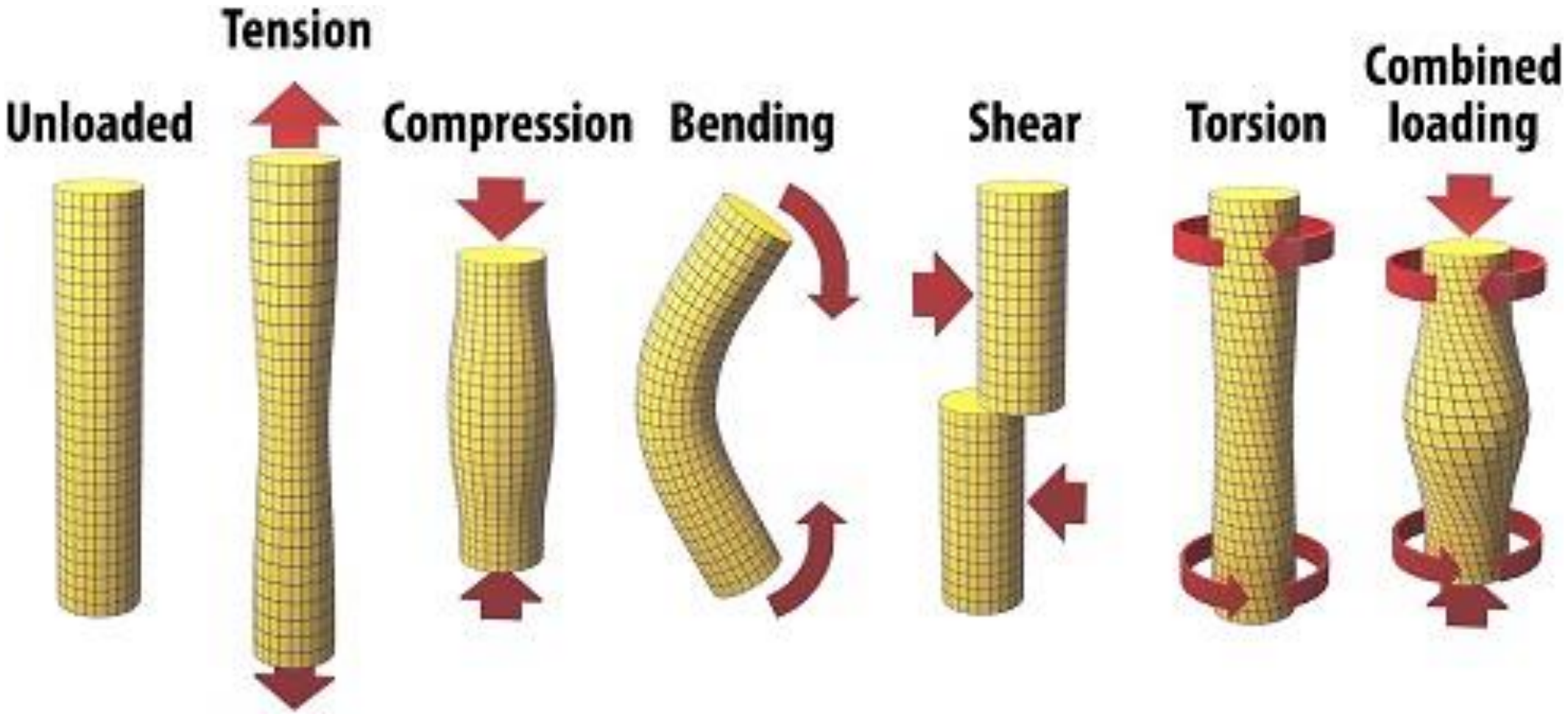


https://youtu.be/XK7NZMZ4YDs

Tensegrity - concept

- https://www.anatomytrains.com/fascia/tensegrity/




Tension
Unloaded
Compression
Bending
Shear
Torsion
Combined loading



Skeletal properties

Ability to resist mechanical load

Quantity
Mass
Areal density

Organization
Geometry
Architecture

Quality
Mineral
Microdamage
Crystal structure
Collagen

- Genetics
- Physical activity history
- Nutrition (caloric intake, calcium, vitamin D)
- Hormonal status (incl. menstrual function)
- Injury history (i.e. recent immobilization, BSI)
- Primary or secondary bone disease



Mechanical load

Loading magnitudes, rates and repetitions

Training features (intensity, duration, frequency)

Muscle (strength, endurance, activation)

Biomechanics (gait mechanics, impact characteristics)

Equipment (shoes, inserts)

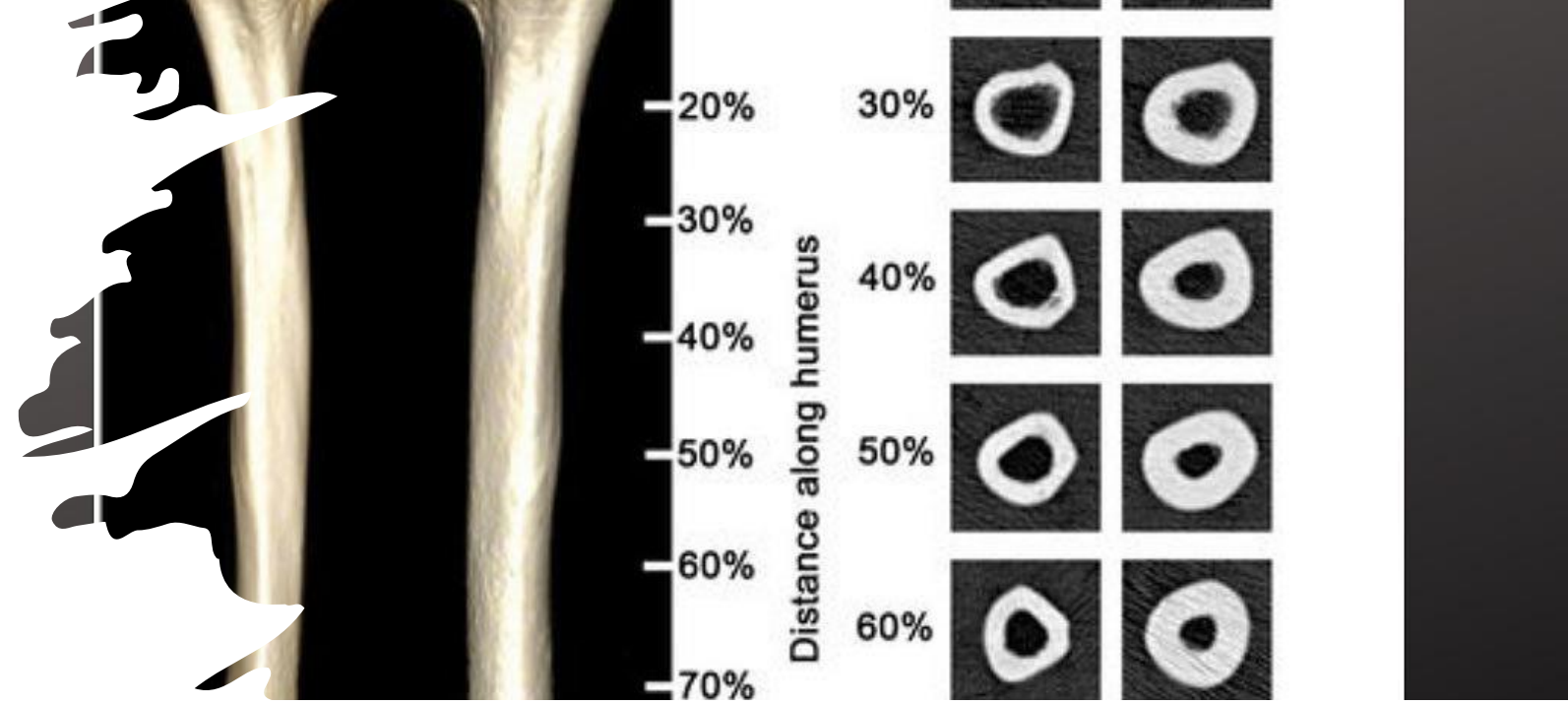
Environment (training surface)

Adaptation

Wolff's law – bones adapt and change according to the stresses that it is subjected to.

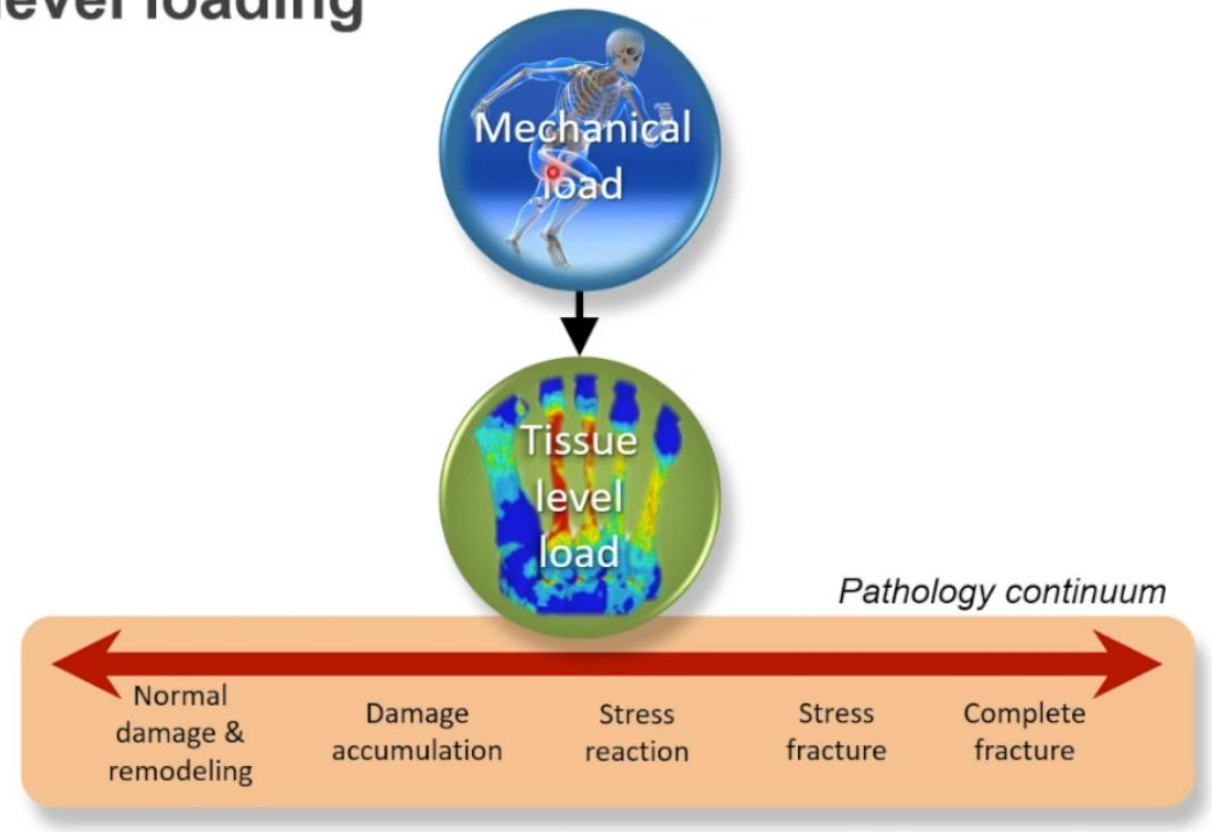
Bones adapt to loading:

The humerus of A major league baseball player's throwing arm is twice as strong.

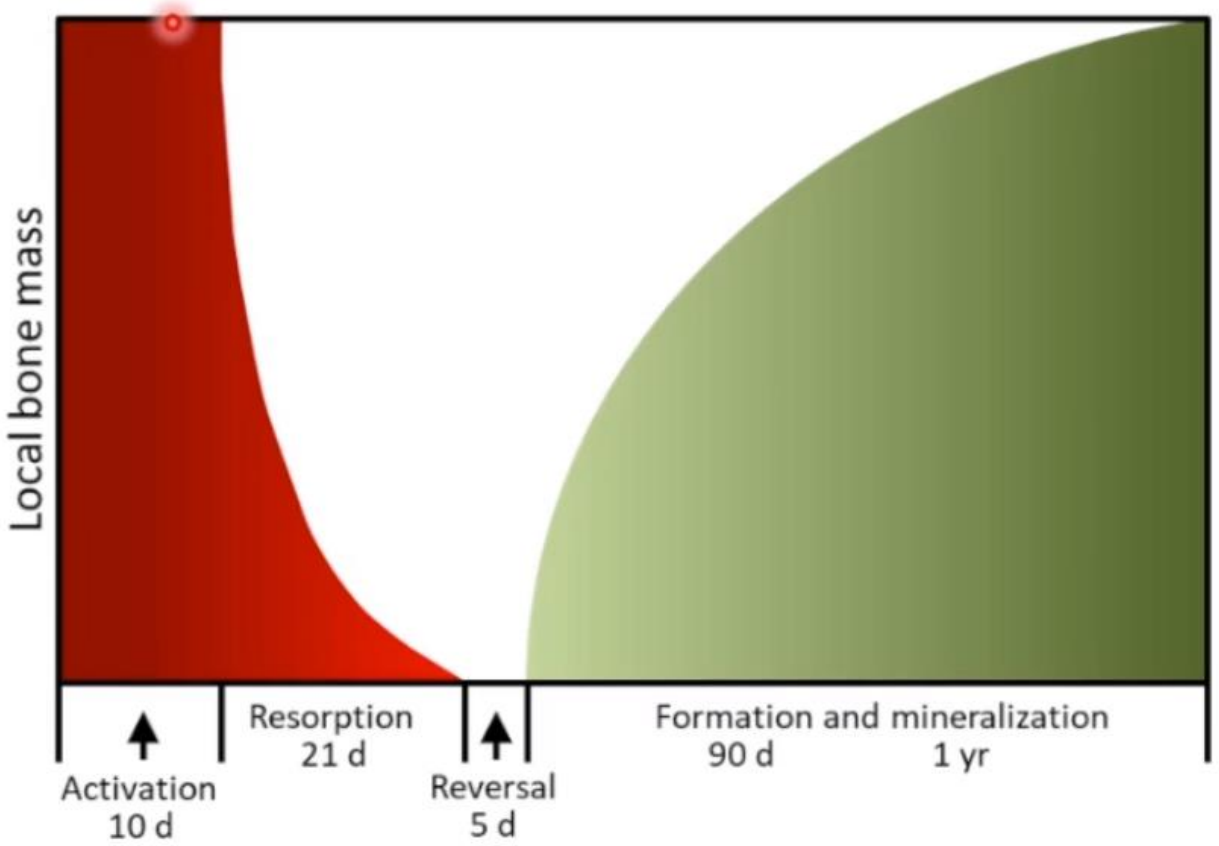


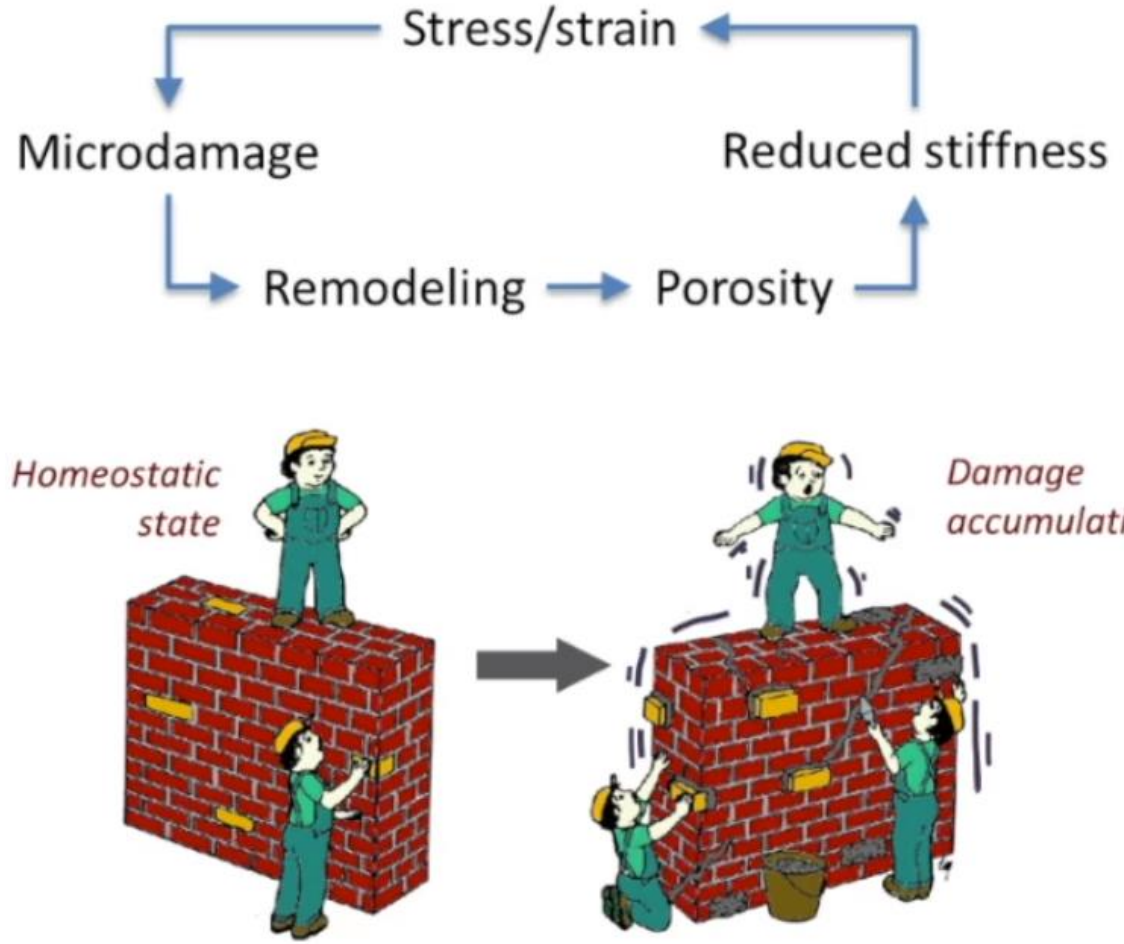


Tissue level loading



Remodeling period and feedforward loop





Bones and force:

- Resist gravity with or without muscle activity – static
- Dynamic forces – resist multi-directional forces
- Intrinsic – ligaments, tendons
- Extrinsic – nonphysiological

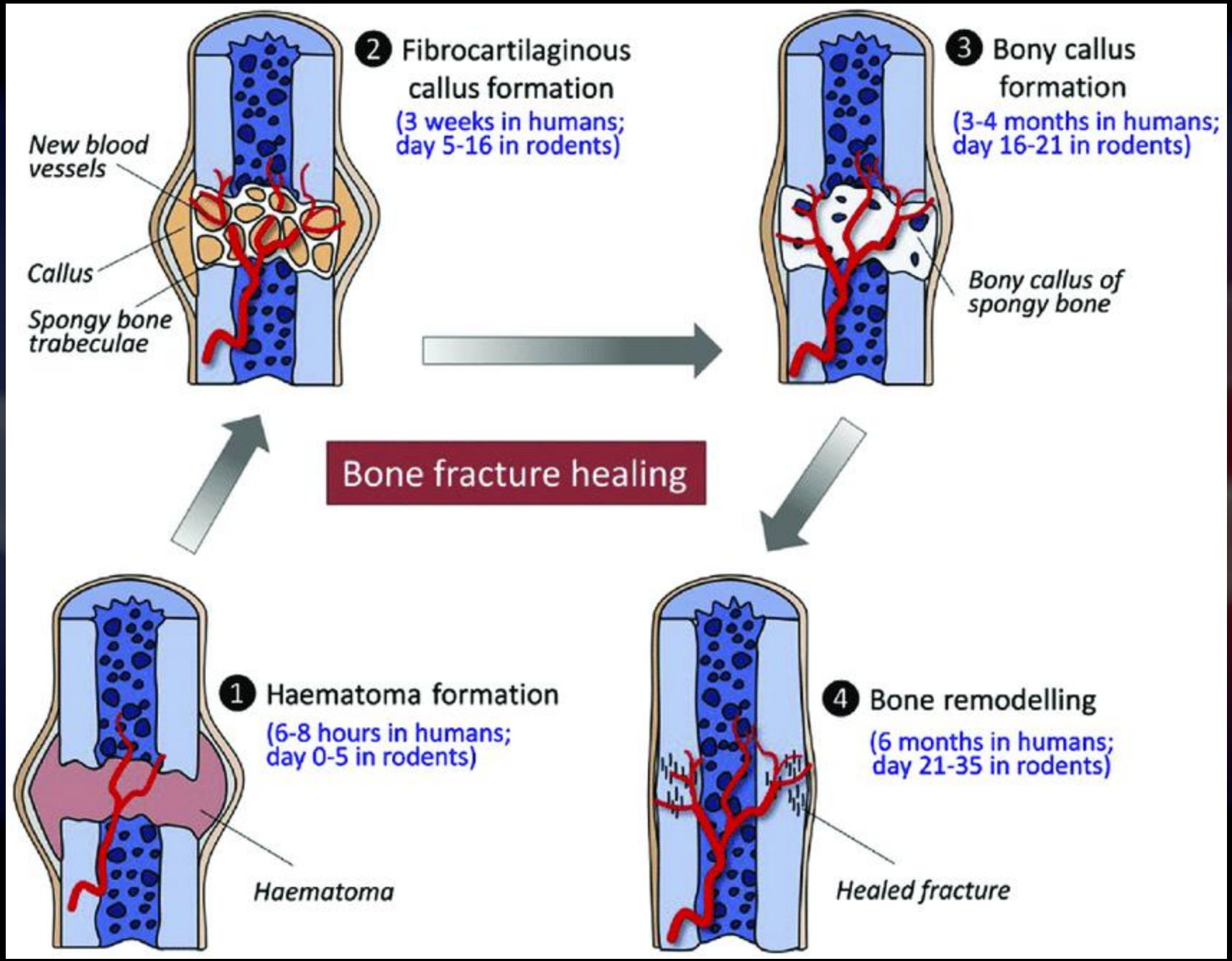
Indirect Fracture Healing

- Secondary – most common
- Endochondral – healing with cartilage
- Intramembranous – healing without cartilage

Can occur with no operation and casting.

Can occur with an operation.

Micro motion and weight-bearing are beneficial, but too much motion or weight-bearing can result in failure.


2 Fibrocartilaginous callus formation
(3 weeks in humans; day 5-16 in rodents)
New blood vessels
Callus
Spongy bone trabeculae
3 Bony callus formation
(3-4 months in humans; day 16-21 in rodents)
Bony callus of spongy bone
Bone fracture healing
1 Haematoma formation
(6-8 hours in humans; day 0-5 in rodents)
Haematoma
4 Bone remodelling
(6 months in humans; day 21-35 in rodents)
Healed fracture

Phase 1: Healing: 24 hours – peak to 7 days post fracture

- Haematoma (hematoma) – Immediate
- Inflammatory Response
  - Angiogenesis
  - Haematoma Coagulation around #ends
  - Fibroblast replication

Granulation – fibroblasts and small blood vessels

Mesenchymal Stem Cells – migrate and differentiate into cells – osteoclasts to reabsorb dead ends of bone.

Phase 2 – Reparative – Soft Callus – peak 7-9 days soft callus, 5-16 days

- Granulations tissue – fibrin
- Soft Callus – fibrin & chondroblasts, hyaline cartilage
- Limited weight-bearing.

Phase 3 – Bony Callus Formation 5-16 days

- Hard Bony Callus at fracture site – 2-4 months
- Hard callus reabsorbs the soft callus



Phase 4 – Remodelling

- Begins 3 – 4 weeks post fracture, lasts for years.
- Hard callus brings stability, but the biomechanical properties of normal bone take longer
- Secondary resorptive phase (hard callus resorption by osteoclasts, and lamellar bone deposition by osteoblasts.

Direct Fracture Healing

Direct healing does not commonly occur in the natural process of fracture healing.

Requires anatomical reduction of the fracture ends, without any gap formation, and a stable fixation.

This is the primary goal to achieve after open reduction and internal fixation surgery.

When these requirements are achieved, direct bone healing can occur by direct remodelling of lamellar bone, the Haversian canals and blood vessels.

Usually takes from a few months to a few years, before complete healing is achieved.

Primary healing of fractures can either occur through contact healing or gap healing.

Both attempt to directly re-establish an anatomically correct and biomechanically competent lamellar bone structure.

Direct bone healing can only occur when an anatomic restoration of the fracture fragments is achieved with rigid fixation.

Bone on one side of the cortex must unite with bone on the other side of the cortex to re-establish mechanical continuity.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3105171/

Cutting cones are formed at the ends of the osteons closest to the fracture site. The cones consist of osteoclasts which cross the fracture line, generating longitudinal cavities at a rate of 50–100 μm/day.

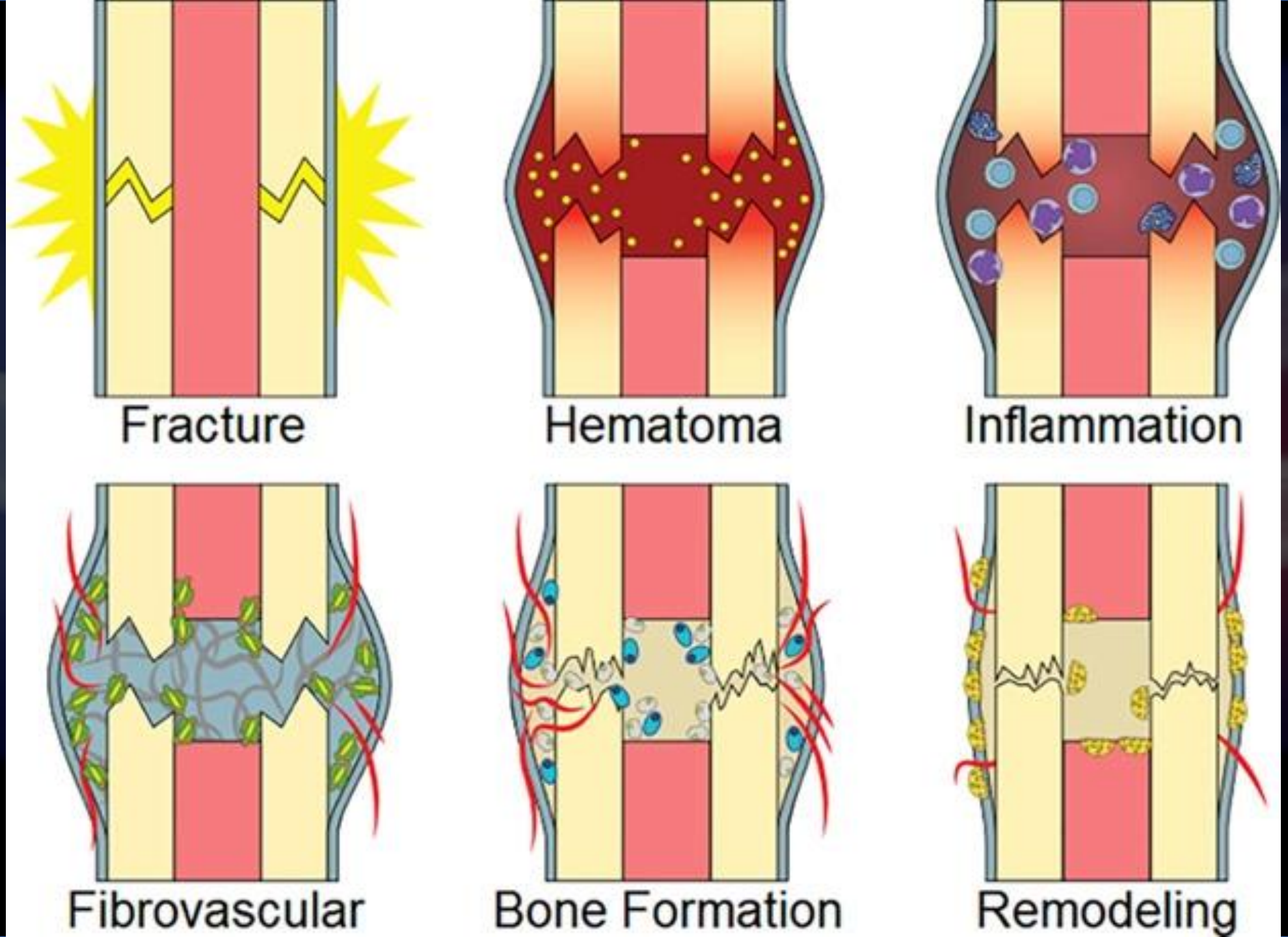
These cavities are later filled by bone produced by osteoblasts residing at the rear of the cutting cone.

This results in the simultaneous generation of a bony union and the restoration of Haversian systems formed in an axial direction.

Re-established Haversian systems allow for penetration of blood vessels carrying osteoblastic precursors.[15, 21]

The bridging osteons later mature by direct remodelling into lamellar bone resulting in fracture healing without the formation of periosteal callus.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3105171/


Fracture
Hematoma
Inflammation
Fibrovascular
Bone Formation
Remodeling

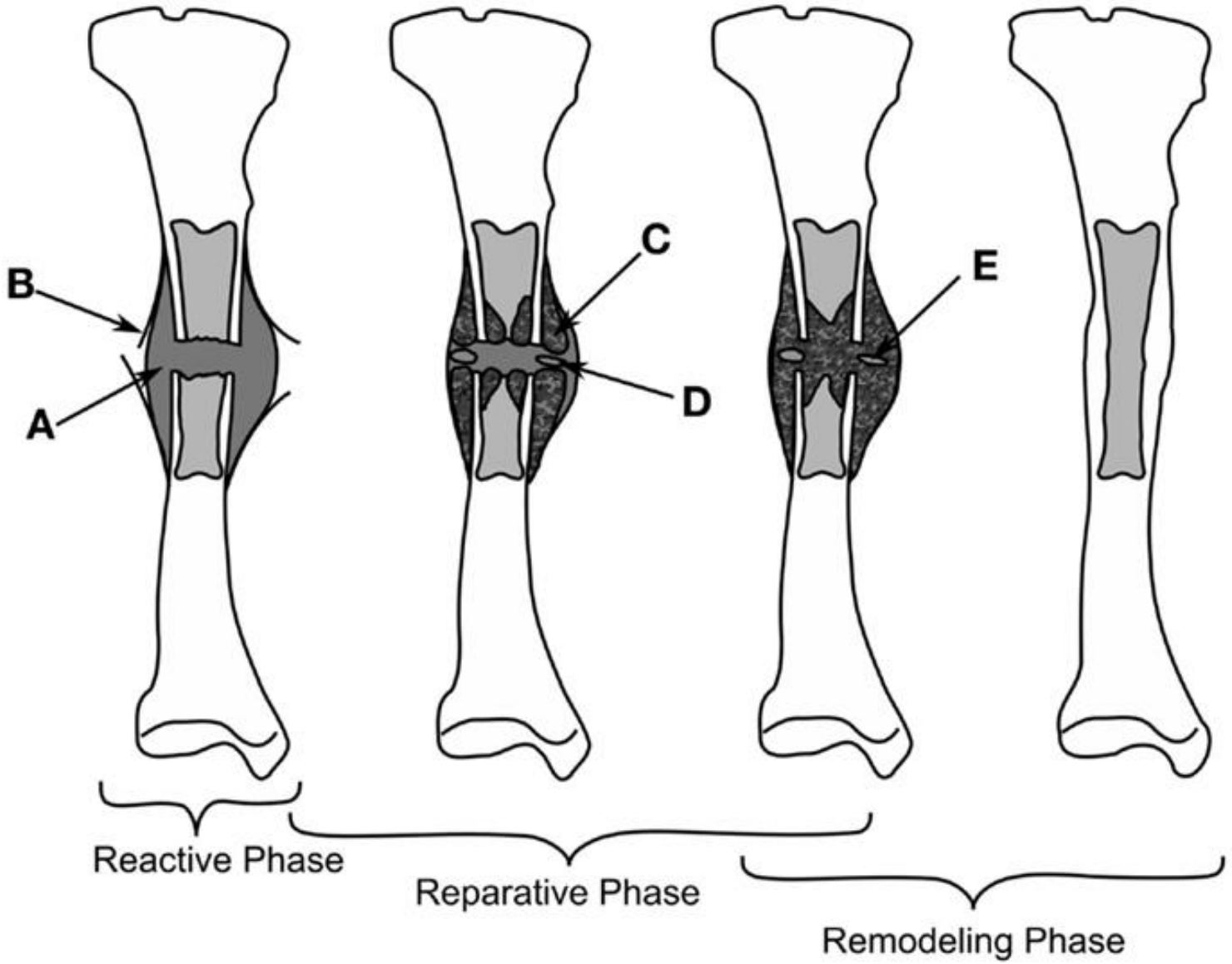
Fractures missed during Emergency Assessment

- Carpal bone injuries
- Occult elbow fractures
- Femoral neck fractures
- Posterior shoulder dislocations
- Epiphyseal plate injuries
- Pubic ramus fractures
- Lisfranc (tarsometatarsal) fractures

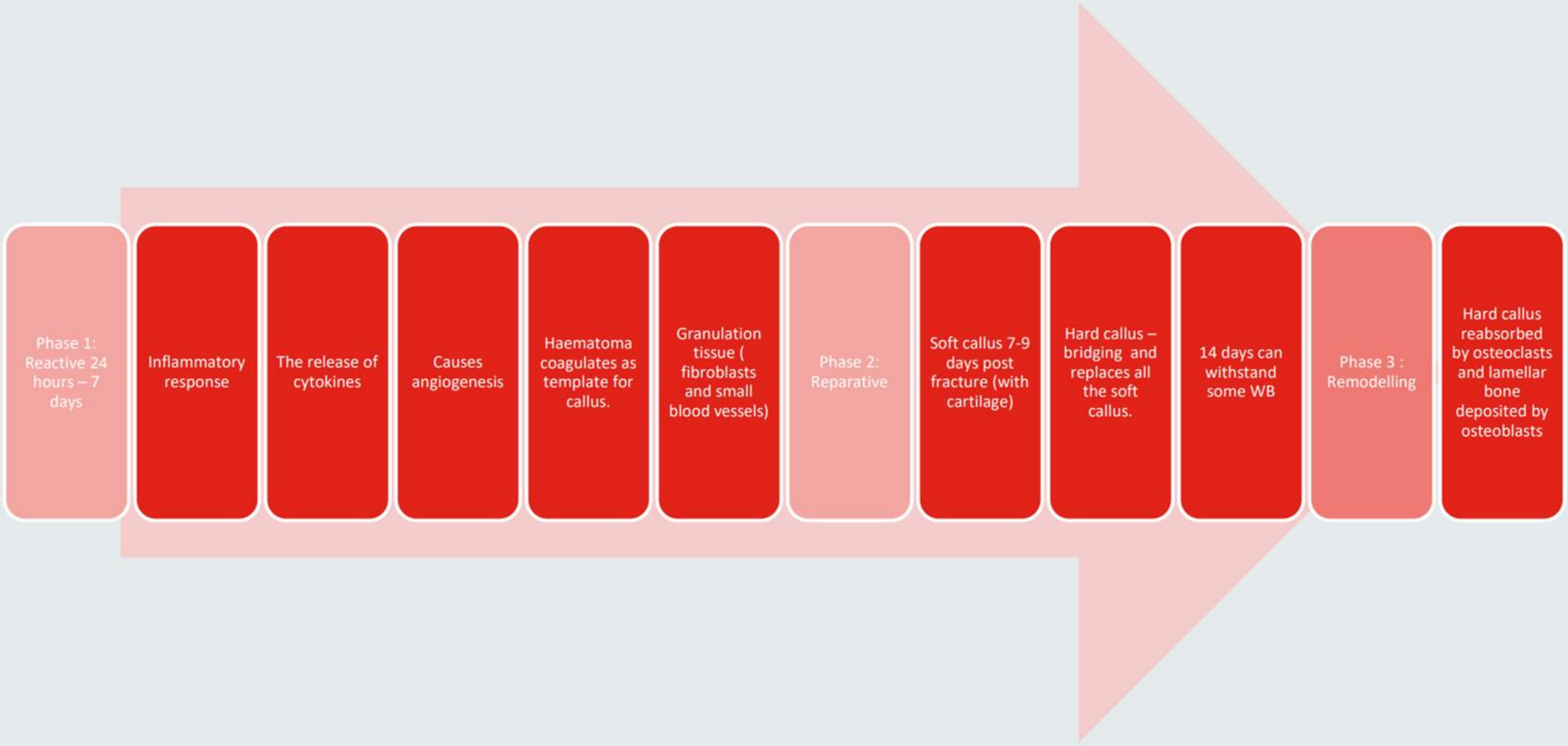
- Magee, David J.. Orthopedic Physical Assessment - E-Book (Orthopedic Physical Assessment (Magee)) . Elsevier Health Sciences. Kindle Edition.

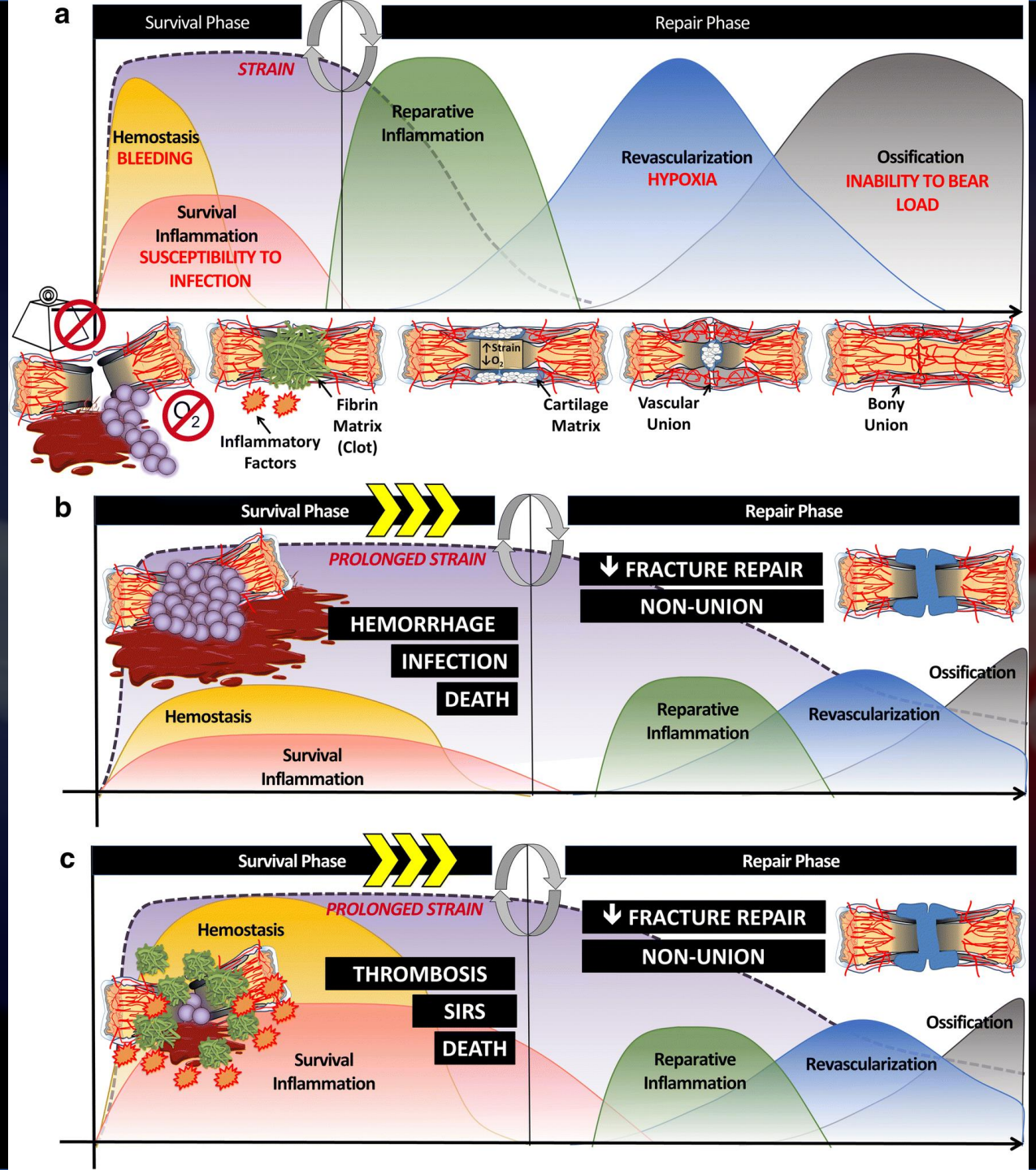
Table 1-4 Pain Descriptions and Related Structures

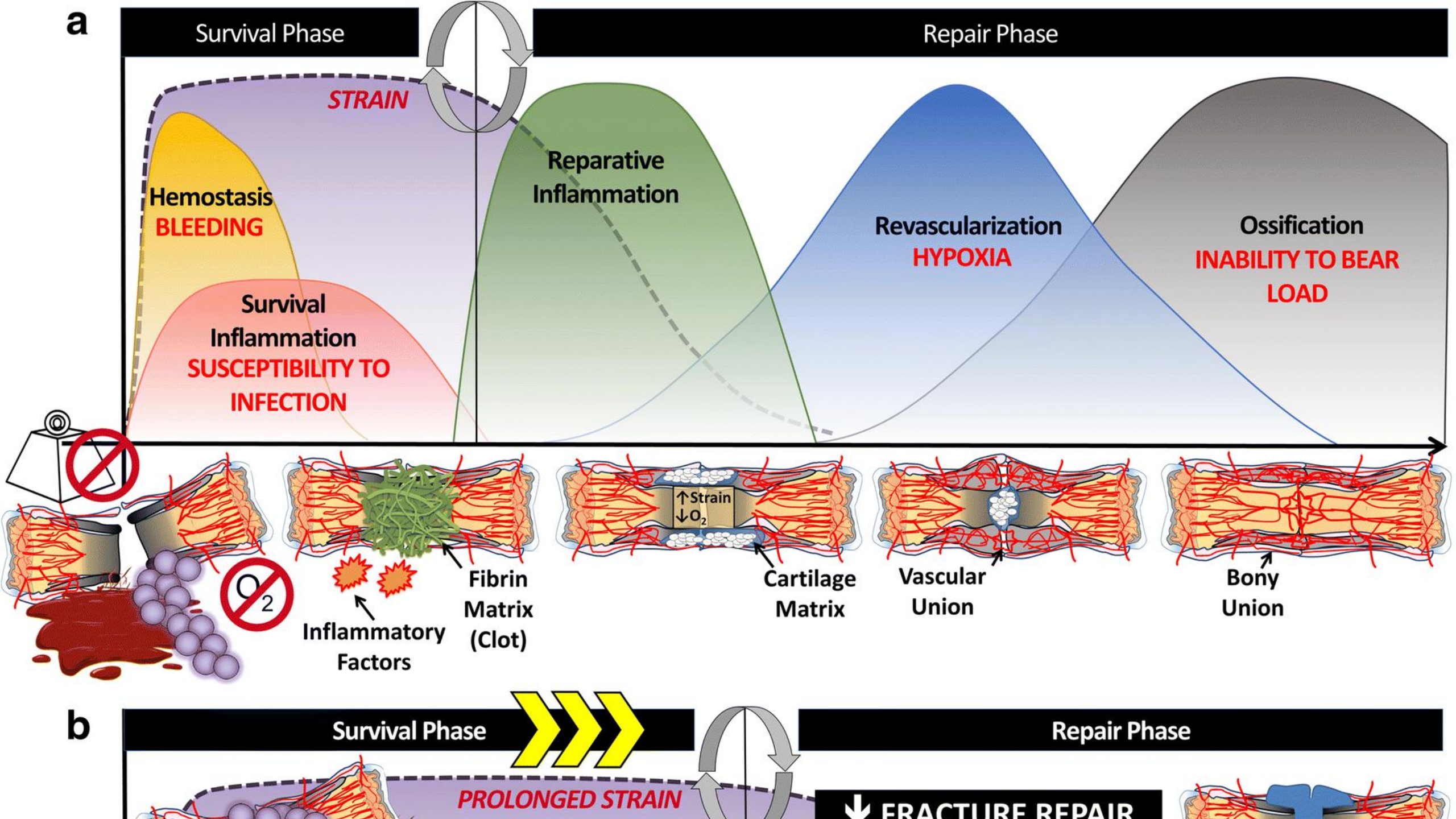
Type of Pain	Structure
Cramping, dull, aching	Muscle
Dull, aching	Ligament, joint capsule
Sharp, shooting	Nerve root
Sharp, bright, lightning-like	Nerve
Burning, pressure-like, stinging, aching	Sympathetic nerve
Deep, nagging, dull	Bone
Sharp, severe, intolerable	Fracture
Throbbing, diffuse	Vasculature

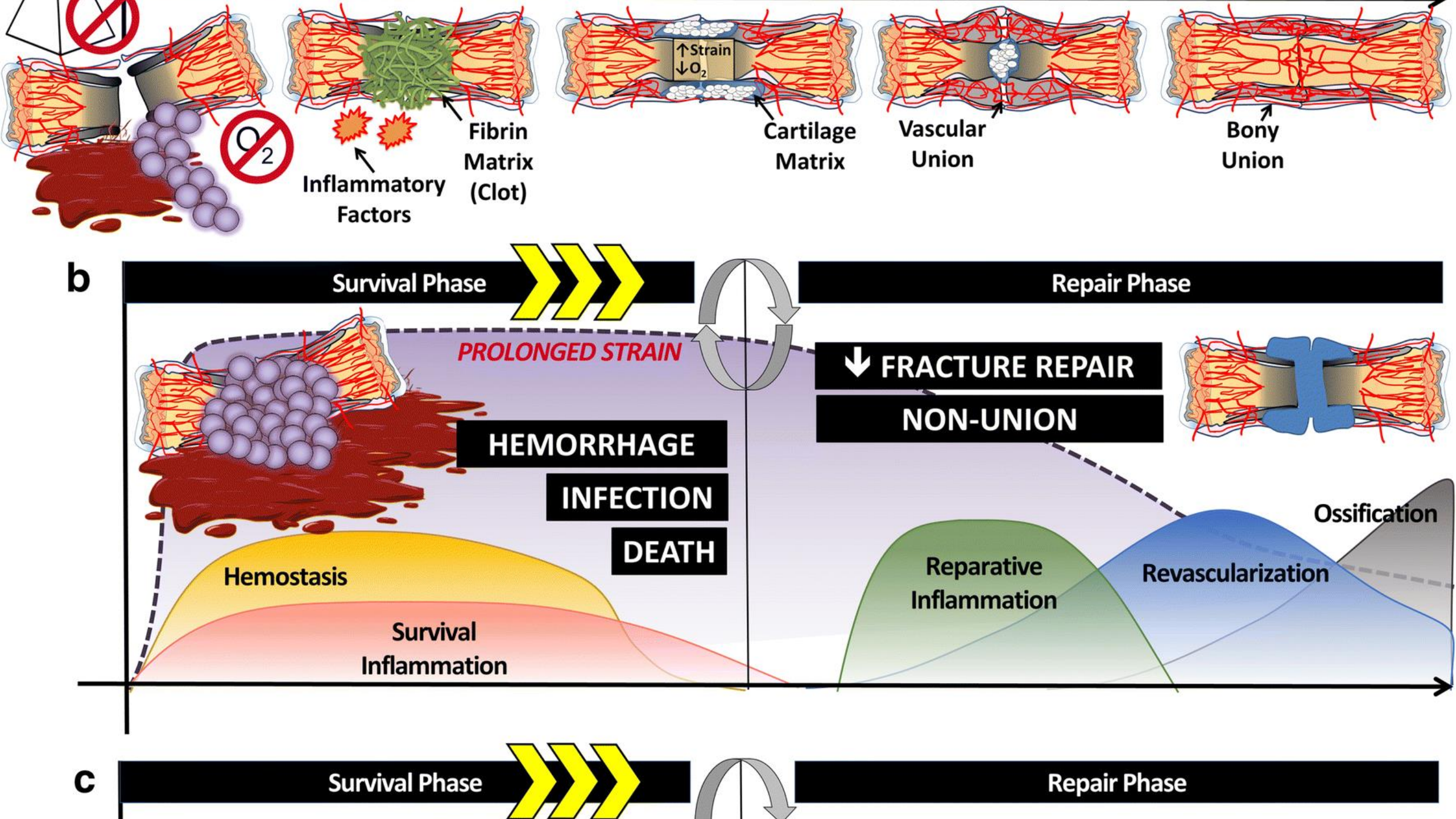


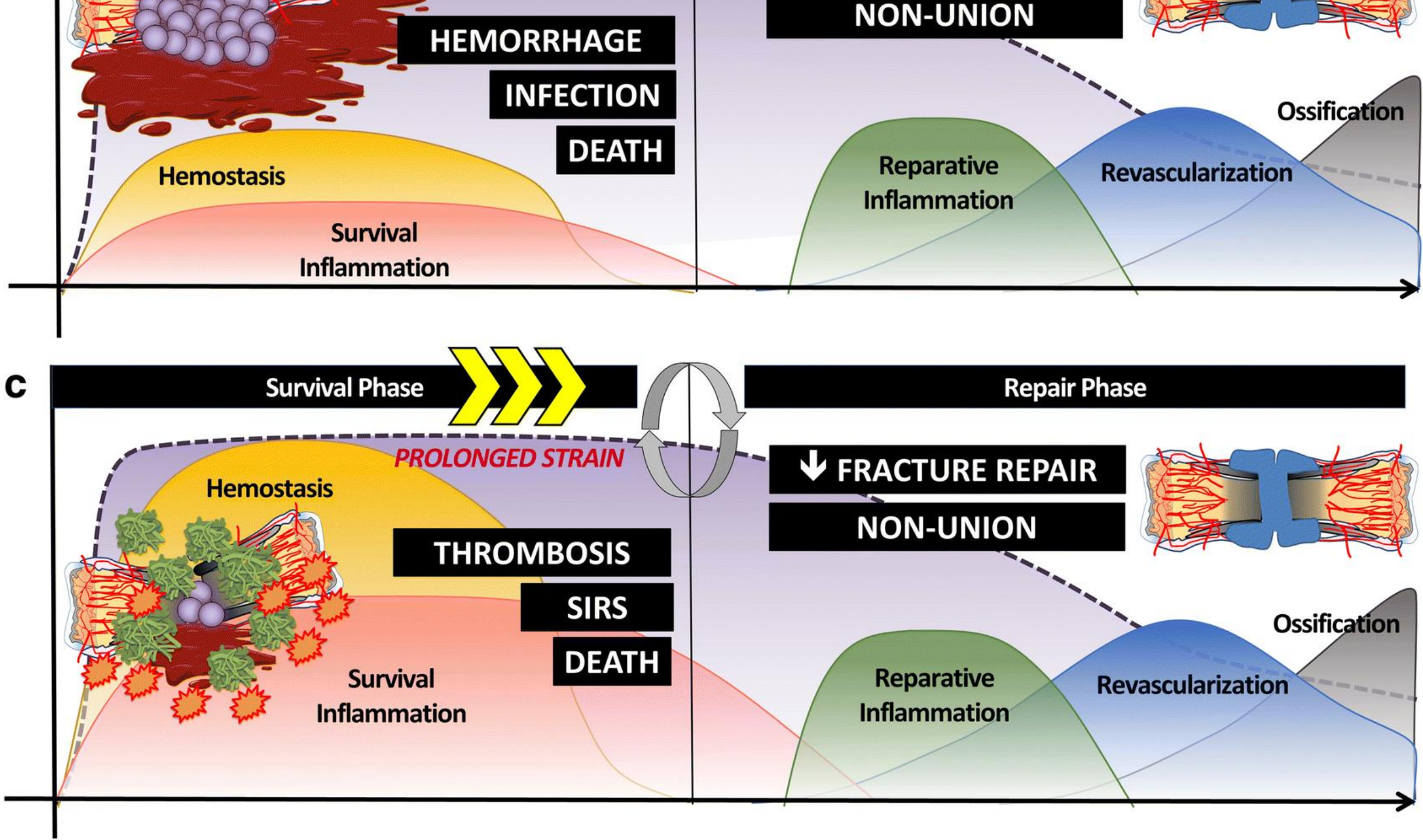
Summary of phases


Phase 1: Reactive 24 hours – 7 days
Inflammatory response
The release of cytokines
Causes angiogenesis
Haematoma coagulates as template for callus.
Granulation tissue (fibroblasts and small blood vessels)
Phase 2: Reparative
Soft callus 7-9 days post fracture (with cartilage)
Hard callus – bridging and replaces all the soft callus.
14 days can withstand some WB
Phase 3 : Remodelling
Hard callus reabsorbed by osteoclasts and lamellar bone deposited by osteoblasts


a
Survival Phase
Repair Phase
STRAIN
Hemostasis
BLEEDING
Survival
Inflammation
SUSCEPTIBILITY TO
INFECTION
Reparative
Inflammation
Revascularization
HYPOXIA
Ossification
INABILITY TO BEAR
LOAD
↑Strain
↓O2
O2
Inflammatory
Factors
Fibrin
Matrix
(Clot)
Cartilage
Matrix
Vascular
Union
Bony
Union
b
Survival Phase
Repair Phase
PROLONGED STRAIN
FRACTURE REPAIR
NON-UNION
HEMORRHAGE
INFECTION
DEATH
Hemostasis
Survival
Inflammation
Reparative
Inflammation
Revascularization
Ossification
c
Survival Phase
Repair Phase
PROLONGED STRAIN
FRACTURE REPAIR
NON-UNION
Hemostasis
THROMBOSIS
SIRS
DEATH
Survival
Inflammation
Reparative
Inflammation
Revascularization
Ossification


a
Survival Phase
Repair Phase
STRAIN
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INABILITY TO BEAR
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↑Strain
↓O2
O2
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Factors
Fibrin
Matrix
(Clot)
Cartilage
Matrix
Vascular
Union
Bony
Union
b
Survival Phase
Repair Phase
PROLONGED STRAIN
FRACTURE REPAIR


Inflammatory Factors
Fibrin Matrix (Clot)
↑Strain
↓O2
Cartilage Matrix
Vascular Union
Bony Union
b
Survival Phase
Repair Phase
PROLONGED STRAIN
FRACTURE REPAIR
NON-UNION
HEMORRHAGE
INFECTION
DEATH
Hemostasis
Survival Inflammation
Reparative Inflammation
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HEMORRHAGE
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Orthopaedic Management

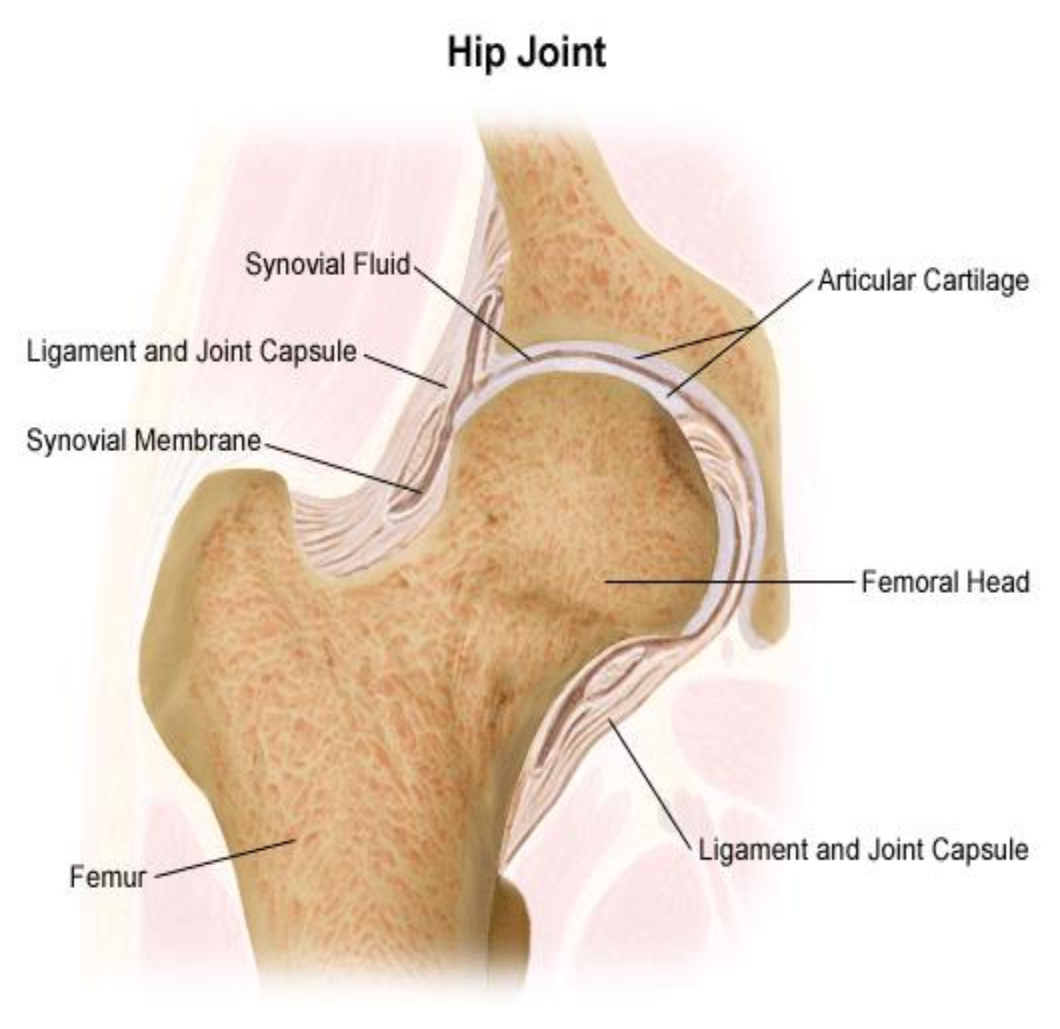
Aims & Objectives

- Provide an outline of basic orthopaedic surgeries
- Describe physiotherapy intervention for various surgical approaches.

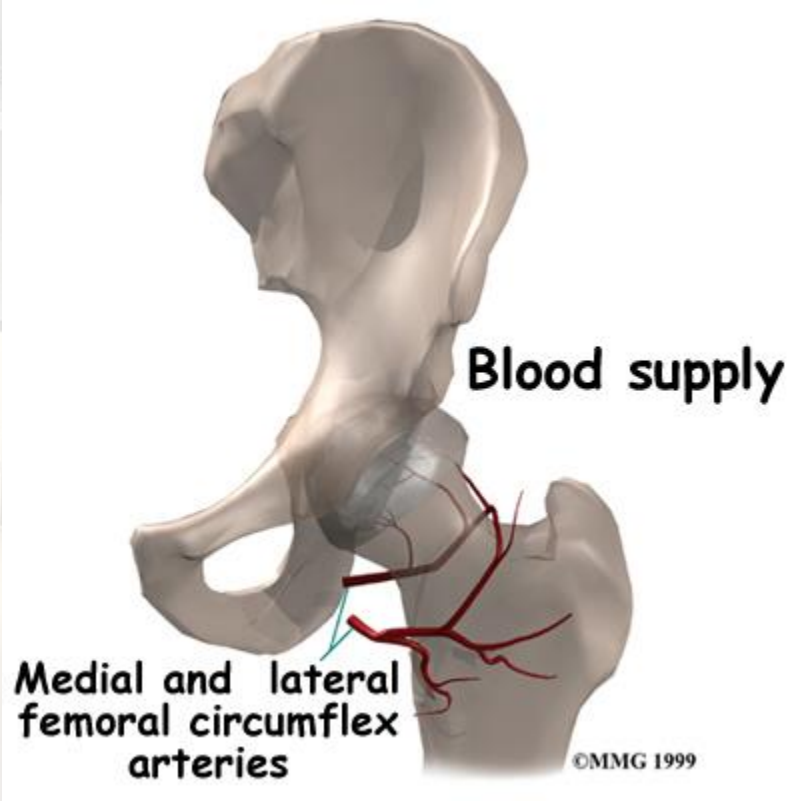
Elective Orthopaedic Hip Surgery



- Total Hip Replacement (THR)
- Hip Arthroscopy
- Peri-acetabular Osteotomies (PAO)



http://www.rush.edu/rumc/images/ei_0244.gif



http://www.eorthopod.com/images/ContentImages/hip/hip_fracture/hip_fracture_anatomy02.jpg

Total Hip Replacement (THR)

- Indications / pathologies:
  - Arthritis (OA / RA)
  - Avascular Necrosis (AVN)
  - Hip fractures
  - Congenital deformities (e.g. CDH, perthes, shallow acetabulum)

- Presentation:
  - Pain
  - Reduced ROM (capsular pattern) and strength
  - Reduced mobility + impaired gait
  - Reduced functional independence
  - Degenerative changes on x-ray


R

- **Surgery:**

1-2 hours

GA +/- spinal epidural / nerve block

8-12 inch antero-lateral incision

Femoral head+surgical neck removed

Acetabulam reamed to anatomical floor

Femur reamed to place femoral stem

Test implants → insert prosthesis +/- cement

Close + drain + dressing





http://www.joint-replacements.com/hipstem.jpg

http://www.jisrf.org/graphics/total_hip_replacementMD.jpg

Physiotherapy NO HIP PRECAUTIONS

Mile stone lead, this is what we aim for:

- **DAY ZERO**
  - **Check Notes**
  - **Check Observations**
  - **Test Sensation**
  - **Quad activation**
  - **Stand patient.**

- **DAY ONE**
  - **Collect Social History**
  - **Teach Bed exercises**
  - **Assess transfers and mobility.**

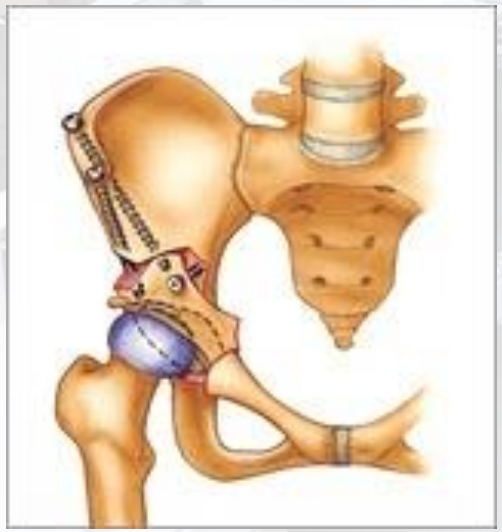
- **DAY TWO**
  - **Progress mobility to elbow crutches**
  - **Teach Standing exercises**
  - **Aim for independence.**
  - **Stairs Ax i**
  - **Aim for Discharge**

- **Each patient is an individual and all have various co-morbidities.**
- **At GSTT we see our patients twice a day.**
- **Average Length of Stay is approx 2 days**

Peri-acetabular Osteotomies

- Indications:

Hip dysplasia – morphological abnormalities of the acetabulum



http://www.capitalortho.com.au/Capital_Orthopaedics/Periacetabular_Osteotomy_files/illustration7.jpg

- Surgery:
  - Open surgery
  - 3 x bony cuts: Ilium, Ischium & Pubis
  - Rotation and re-alignment of acetabular complex to improve joint articulation
  - 3-4 screw fixation in new position





25 year old female with right hip dysplasia treated successfully with peri-acetabular osteotomy (PAO).

http://www.thewellingtonhipunit.com/images/hip_AdultHip.jpg

- Acetabulum is shallow and upward sloping
- Causing groin pain, catching / impingement, instability, early wear
- Surgical cut to pelvis, the acetabulum is rotated to create greater coverage.
- Result = reduction in symptoms and prevention of accelerated OA

- Post-op:
- Exercise programme/ active assisted ROM
- Functional re-education: transfers
- Mobility Ax (TWB/ PWB- 20kg)
- gait re-ed
- Advice:
  - Pain Control
  - Referral to Hydrotherapy whilst patient is TWB/PWB

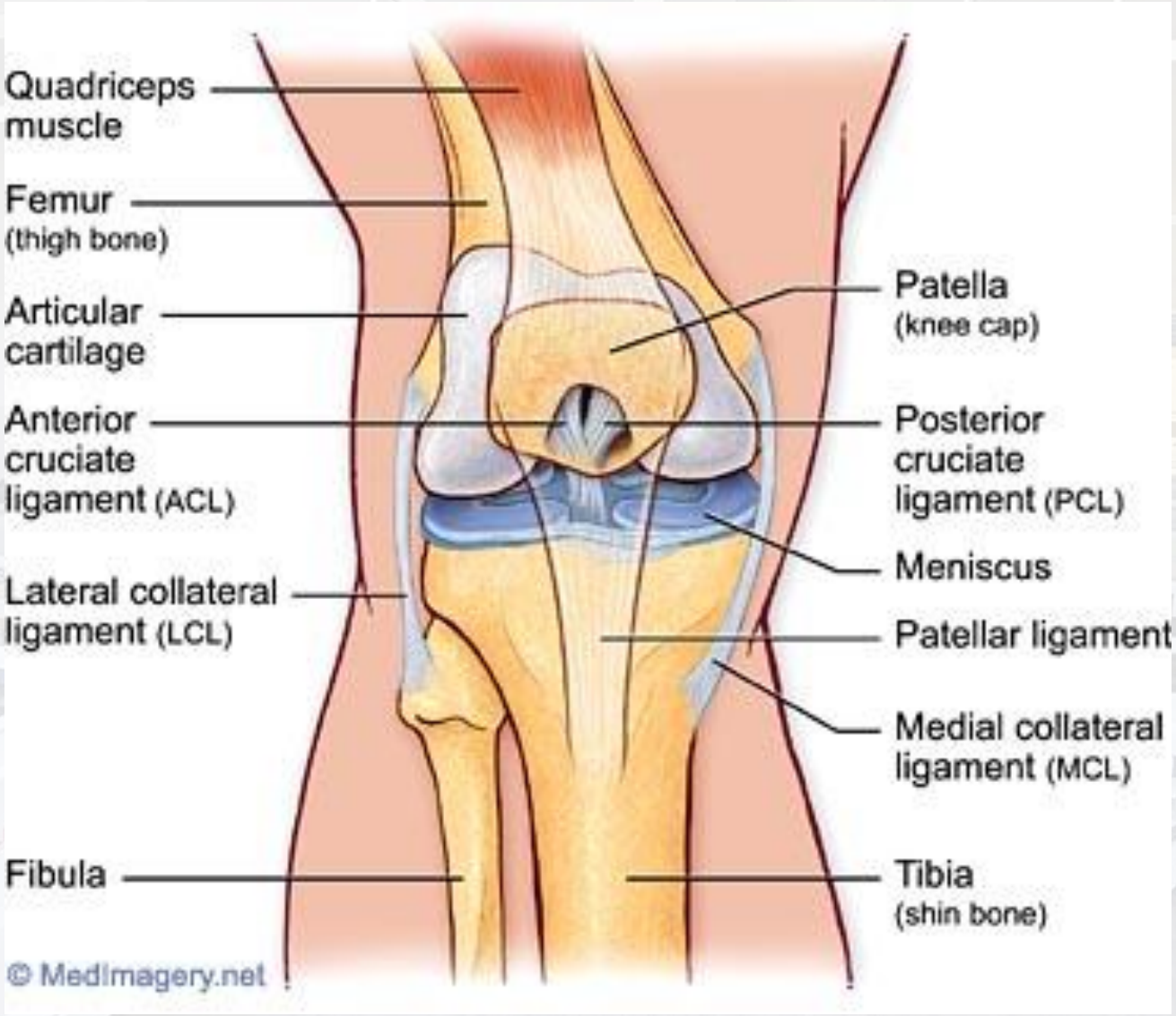
Weight Bearing Status

- Important to check the WBS of all lower limb orthopaedic surgery- over load the joint or fracture site can lead to delayed healing, non-union or breakdown in wound.
- NWB-
- TWB-
- PWB-
- FWB-

Elective Orthopaedic Knee Surgery

- Uni-compartmental knee replacement (UKR)
- Total Knee Replacement (TKR)
- Knee arthroscopy
- ACL Reconstruction

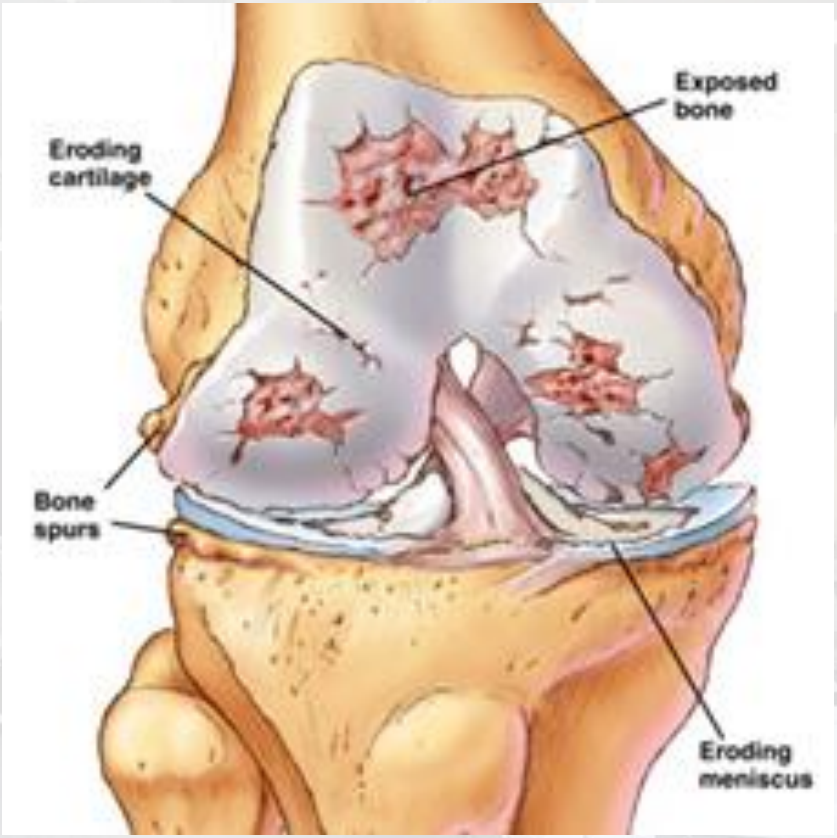




http://www.orthspec.com/drawings/about_the_knee--01.jpg



http://www.yorkshirekneeclinic.co.uk/images/D3.jpg



http://www.zimmer.co.nz/web/images/patient_education/OsteoarthritisKnee1.jpg

Uni-compartmental (UKR)

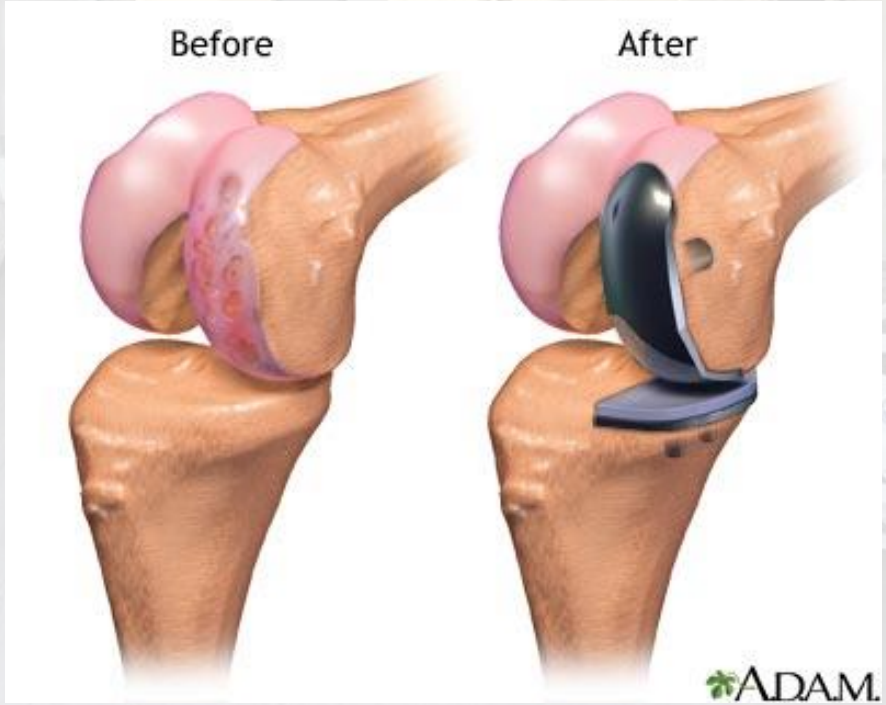
Indications:

Single compartment OA

Younger, more active patients

Surgery:

Replacement of medial > lateral femoral condyle, tibial insert and polyethylene liner



http://assets.aarp.org/external_sites/adam/graphics/images/en/21769.jpg

Total Knee replacement (TKR)

- Indications:

OA

RA

Trauma

Avascular Necrosis

Infection

Ca / tumour

- Presentation:

Swelling

Deformity (varus / valgus)

Weakness (quads)

Stiffness (flex + ext)

Pain (medial joint line)

Degenerative changes on x-ray

Reduced mobility and function



http://www.yorkshirekneeclinic.co.uk/images/D5.jpg

<u>Surgery</u>:

Anterior incision

Patella displaced

Femoral and tibial articulating surface cut.

Test Implants

Implant and Polyethylene insert.

Close.

Bandage



http://www.kneepaininfo.com/pics/oatotalknee.jpg

Physiotherapy

Mile stone lead, this is what we aim for:

- **DAY ZERO**
  - **Check Notes**
  - **Check Observations**
  - **Test Sensation**
  - **Quad activation**
  - **Stand patient.**

- **DAY ONE**
  - **Collect Social History**
  - **Teach Bed exercises**
  - **Assess transfers and mobility.**
  - **ICE therapy- Cryocuff**
  - **Check Knee Flexion & Extension ROM**

- **DAY TWO**
  - **Progress mobility to elbow crutches**
  - **Teach Standing exercises**
  - **Aim for independence.**
  - **Aim to achieve 0-90 AROM**

- **DAY THREE**
  - **Aim for independent mobility with crutches**
  - **Aim independent with stairs**
  - **All patients to have out-patients Physiotherapy to improve ROM and strength..**

Emphasis the importance of active range of movement.
PUSH FOR ROM!!!

Knee arthroscopy ("Keyhole Surgery")

- Indications:

Examination under anaesthetic (EUA)

Aids diagnosis and assists surgical intervention

Pain and early degenerative conditions



http://www.kneeclinic.info/images/knee-arthroscopy.jpg

- Surgery:

GA / local

2-3 portals

Saline pumped into joint space to expand and expose joint

Arhroscope inserted to view intra-articular space

Instrumentation for surgical intervention

- Surgeries:
  - Biopsy
  - Washout
  - Debridement
  - Menisectomy
  - Meniscal sutures / repair
  - Ligament repairs & reconstruction
  - Micro-fracture

- Post-op physio for routine arthroscopy:
  - Mobilise FWB with EC
  - Encourage full AROM and strength with ex.regime
  - Complete stairs assessment

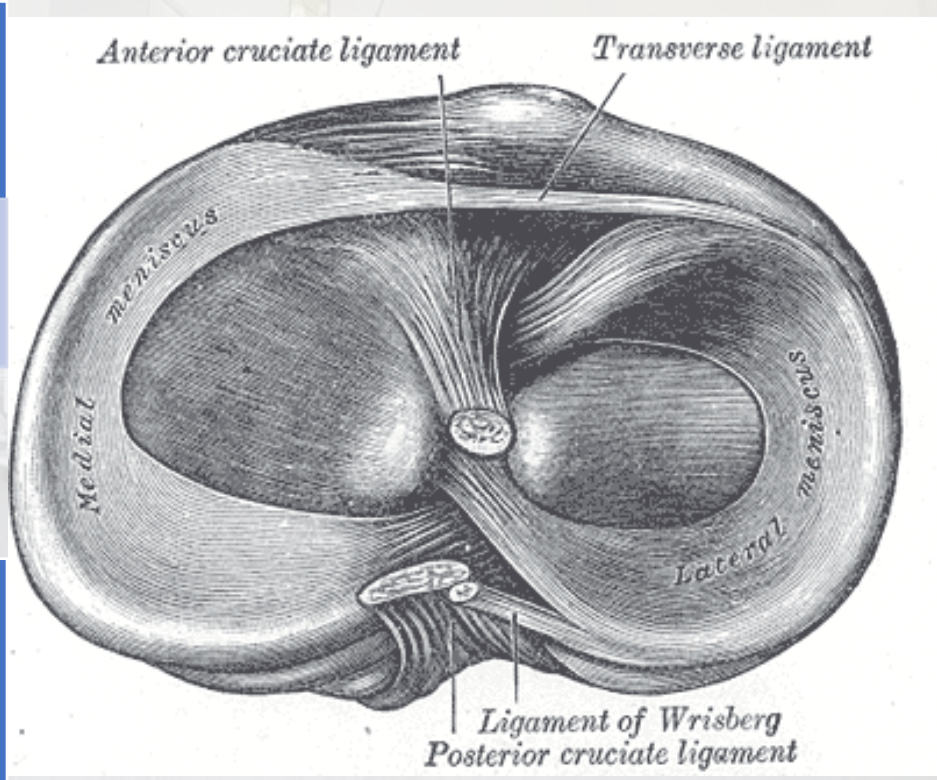
MORE COMPLEX ARTHROSCOPIC PROCEDURE WILL FOLLOW DIFFERENT PROTOCOLS!

Anterior Cruciate Ligament (ACL)

Attachments:

Antero-med tibial plateau to postero-lat femoral intercondylar notch.

Role of ACL: To prevent forward translation of the tibia.



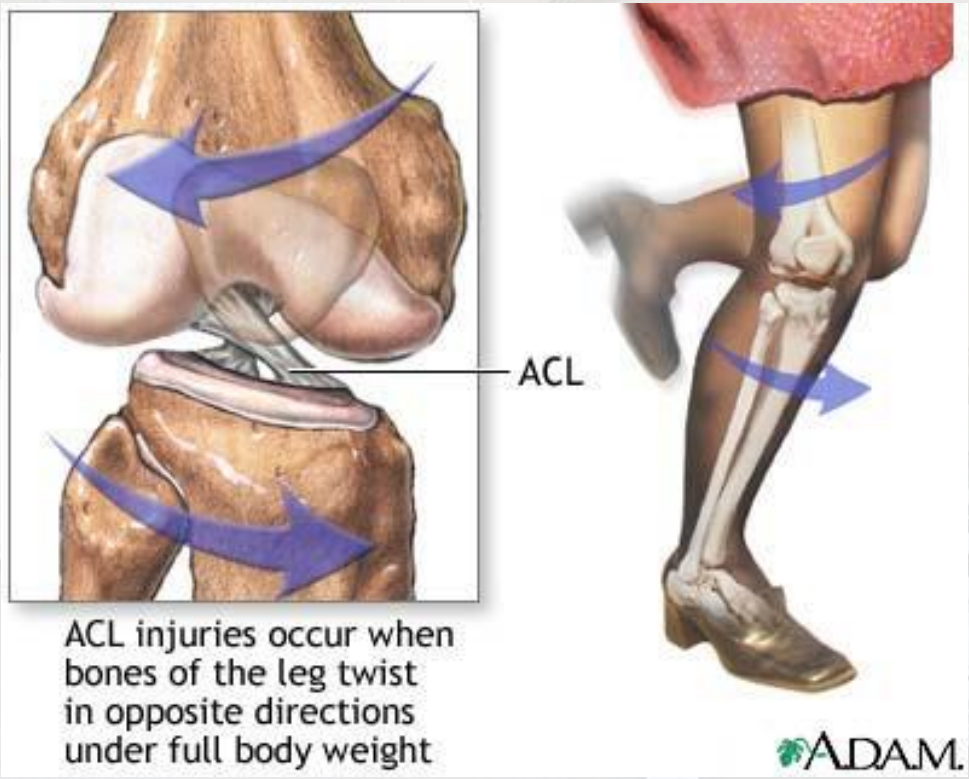
http://upload.wikimedia.org/wikipedia/commons/a/a8/Gray349.png

ACL - rupture

Mechanism of injury:

Foot fixed, flexion and rotation under load.	Can be some involvement of collateral ligaments and menisci

Diagnostics: Lachmann's and anterior draw + MRI



http://z.about.com/d/p/440/e/f/18003.jpg

ACLR

- <u>Operation:</u>

2 options for graft –

i) Bone-patella- tendon-bone (BPTB)

ii) Hamstring

Screw fixation / absorbable sutured into place

Brace to maintain extension for 48 hrs.



- <u>Post-op physio:</u>

- RICE
- Advice on analgesia
- Mobilise with EC (FWB / PWB)
- Exercise regime
- Onward referral to physio OPD for ACL rehab.

Prehab is the new Rehab!

- Knee and hip classes.
- Must attend classes prior to surgery.
- Intervention proven to enhance:
  - Patient satisfaction
  - Recovery
  - Manage expectations
  - Reduce anxiety
  - Reduce Length of Stay.

Components to Pre-Hab

- Information about their surgery, length of stay, ward environment etc.
- Information about symptoms.
- Exercise regime to increase strength and flexibility
- Practice using elbow crutches + stairs technique
- OT education re: ADLS and functional transfers (e.g. car)
- Opportunity to ask questions.

Discharge Planning

- “Patients with complex care needs are to be referred to the Discharge Team and Social Services at the earliest opportunity”
- Commitment to a multidisciplinary approach to discharge planning ensures early referral and assessment via weekly multidisciplinary meetings.
- This provides effective communication about the patient’s abilities and needs for care both in hospital and in the community.

Options available

- Outpatients.
  - Independent
  - Improve ROM, strength and gait re-education.
  - Return Back to Sport
- Home-Based Rehab
  - Requiring assist of 1 for:
    - Transfers
    - Mobility ie still on frame
    - Unable/Difficulty with stairs.

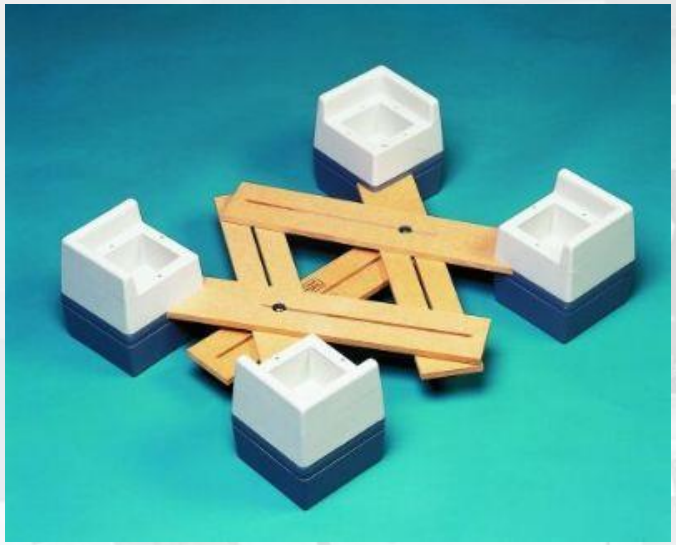
Options

- Bed Based Rehab.
  - Assistance of 2 to transfer
  - Assistance of 2 to mobilise

- Community
  - Ongoing rehab goal
  - Balance Work
  - Falls prevention
  - Outdoor mobility

Functionally independent

Occupational Therapy input











http://www.benefitsnowshop.co.uk/media/hr/images/detail/7215_AA3366.jpg

http://www.carterchemist.co.uk/assets/images/Long_handled_back_sponge.jpg

http://www.assistireland.ie/uploadedfiles/Product_Images/Healthcare_Products/Medication_Products/Toe%20Cleaner%205764.jpg

http://ecx.images-amazon.com/images/I/41KkfuR6cCL._SL500_AA280_.jpg

http://shop.bupa.co.uk/images/product_details/2446/

http://www.medisave.co.uk/images/helping_hand_shoe_helper1.jpg



Crutches

Devices which are used to decrease weight on the lower limb/s.

Pre requisites for crutches

- Good strength of upper limb muscles is required.
- Range of motion of upper limb should be good.
- Muscle group which should be strong is given below -
  - Shoulder flexor, extensors and depressor
  - Shoulder adductors
  - Elbow and wrist extensors
  - Finger flexors











